MSc Health Data Science and Statistics

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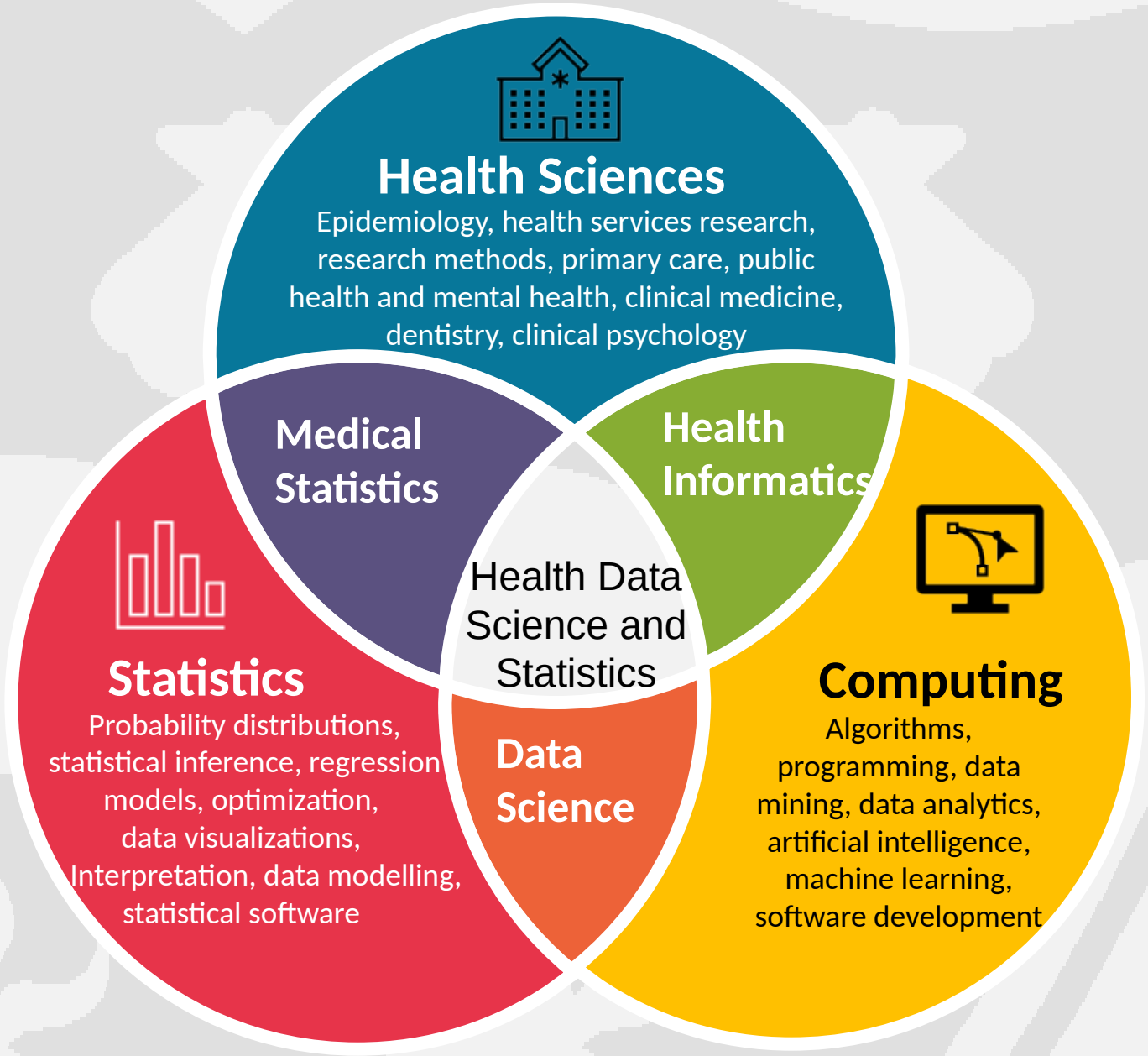
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Room 04, 2-5 Kirkby Place



MSc Health Data Science and Statistics

Overview: This interdisciplinary programme provides you with up-to-date knowledge on data analytics, medical statistics, machine learning and artificial intelligence, as well as the design and analysis of health studies.



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Careers with this subject

- Data analyst
- Data engineer
- Data manager
- Data scientist
- Statistician
- Researcher

- NHS
- Clinical Trials Units
- Pharmaceutical companies
- Biotechnological industries
- Government agencies
- Research institutions
- Consultancy units

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MSc Health Data Science and Statistics: Outline Structure – 20 credits per taught module

	Compulsory modules		Elective modules	
Semester 1	Health Data and Medical Statistics	Big Data Visualisation and Analytics	Systematic Reviews	Software development and databases Information Security Management and Governance
Semester 2	Machine Learning and Artificial Intelligence for Healthcare	Modelling and Analytics for Data Science	Applied Quantitative Research Methods	Computer Vision and Deep Learning Natural Language Processing and Large Language Models

Summer: MSc Project, **60 credits**

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Semester 1: MATH515 Health Data and Medical Statistics

This module introduces health data and medical studies, and related ethical considerations. It discusses statistical techniques for the design, analysis and interpretation of medical studies. Up-to-date professional software for analysing medical studies is studied. Experience in preparing a professional statistical analysis plan is provided.

Aims

- To introduce health data and medical studies, and related ethical considerations.
- To give students an understanding of the design and analysis of medical studies
- To supply experience in preparing a statistical analysis plan.

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Semester 1: MATH517 Big Data Visualisation and Analytics

Sophisticated analytics techniques are needed to visualize today's increasing quantities of Big Data. Up-to-date R tools including dplyr for data manipulation, ggplot2 for visualization, and knitr/LaTeX for document presentation are studied. The module provides considerable experience of writing professionally documented R code using RStudio.

Aims:

- To develop a working knowledge of R using RStudio to develop professionally documented code for Data Science applications.
- To illustrate modern techniques for extracting and presenting strategically useful
- information from a range of data sources including data bases using dplyr and dbplyr, ggplot2,
- knitr and LaTeX tools to produce and present results.

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Semester 2: MATH516 Machine Learning and Artificial Intelligence for Healthcare

This module aims to introduce the cutting-edge techniques in machine learning and artificial intelligence for health data analytics. The practical implementation will be achieved using appropriate software.

Aims:

- To discuss the techniques for machine learning and artificial intelligence as well as their ethical issues in the context of health data
- To supply an introduction to computational techniques using appropriate software in the context of health data analytics.
- To provide experience in presenting the results of a state-of-the-art health data analyses to technical and non-technical audiences.

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Semester 2: MATH518 Applied Data Modelling and Artificial Intelligence

We'll teach you state-of-the-art modelling and analytics techniques for Data Science, so that you can provide informed strategic advice in a broad range of business and related situations. You'll meet techniques to extract knowledge from data and to update that knowledge when new observations become available. We'll also discuss popular Machine Learning algorithms for making classification decisions.

Aims:

- To give students an in-depth understanding of contemporary modelling and analytics techniques for Data Science.
- To provide students with broad tool boxes for frequentist and Bayesian modelling and machine learning.
- To supply employment-relevant problem solving experience.

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Elective modules

Option module 1:

MCR706 Systematic Review

COMP5000 Software Development and Data Bases

COMP5006 Information Security Management & Governance

Option module 2:

MCR702 Applied Quantitative Research Methods

COMP5019 Natural Language Processing and Large Language Models

ROCO510 Computer Vision and Deep Learning

Choose one from option module 1

Choose one from option module 2

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Option Module 1

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Option Module 1- COMP5000: Software development and database

Dr Craig McNeile (craig.mcneile@plymouth.ac.uk), **SECaM**

- Requirements Analysis: Identifying the business needs.
- Relational Model: Identify entities and normalise data to derive the database tables.
- Database Development with SQL: Develop a database based on the results of the analysis using SQL.
- Introduction to python
- Data cleaning with python including using regular expressions and PANDAS.
- Introduction to object oriented python.
- Using python to create to Graphical User Interfaces



Option Module 1 - COMP5006: Information Security Management and Governance

Dr Rory Hopcraft (rory.hopcraft@plymouth.ac.uk), **SECaM**

This module looks at the issues surrounding the management and governance of information security within an organisational context. Consideration is given to the need for related policy, analysis of risk, and the management of organisational assets. Coverage also includes legal and personnel aspects of security, giving an overview of the wide range of laws and regulations governing systems & information security.

Aims:

- To impart an understanding of the underlying principles associated with security management and governance.
- To develop an understanding of security threats and vulnerabilities within modern organizational environments
- To gain an appreciation of the principles and practices of risk analysis as applied to IT systems.
- To provide a broad contextual overview of the law and regulations relating to information systems security and the need for legislative compliance.



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Option Module 1 - MCR706: Systematic Review

Mr Keith Walker (keith.walker@plymouth.ac.uk), **School of Health Professions**

- Examine the nature of evidence and its role in healthcare to further understand how evidence transfers into health care practice.
- Methods and tools to appraise and synthesize evidence from research literature and documentary sources.
- Hands-on experience using JBI software tools to draft a systematic review protocol, including
- Identifying a question, Searching, retrieving, Selecting research for the review.
- Analyse sample quantitative and qualitative research and expert opinion papers as part of the review process and use software to perform a meta-analysis and meta-synthesis of selected studies

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Option Module 2

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Option module 2: ROCO510 Computer Vision and Deep Learning

ROCO510

Computer Vision and Deep Learning 20 credits

This module provides advanced knowledge of computer vision systems and state-of-the-art deep learning techniques. It covers the theoretical foundations of visual systems, including feature detection, recognition, segmentation, stereo vision, and calibration. Learners will explore robust AI-driven deep learning methods, gaining both theoretical insights and practical skills for computer vision applications.



50% Coursework



50% Examinations

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Option Module 2 - COMP5019 Natural Language Processing and Large Language Models

Discover how machines understand and generate language. From syntactic analysis to transformer architectures, this module explores modern NLP and large language models. You'll build and evaluate systems that can translate, summarise, answer questions and shape the future of communication.

Short Module Description

Natural Language Processing (NLP) and Large Language Models (LLMs) form the backbone of many modern AI systems, including chatbots, search engines, and intelligent assistants. This module introduces the core principles of NLP alongside state-of-the-art LLMs such as BERT and GPT. Students will explore both theoretical foundations and practical techniques, gaining hands-on experience in building, fine-tuning, and evaluating language-based AI systems for real-world applications.

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Option Module 2 - MCR702: Applied Quantitative Research Methods

Mrs Munira Khan (munira.khan@plymouth.ac.uk), **School of Health Professions**

This module will allow students to critically appraise specific quantitative methods to address research questions. It will provide them with practical experience of using quantitative data analysis techniques. The methods reviewed will include:

- randomised controlled trials, cross-over designs, cohort, and case control studies.
- Data analysis techniques covered will include assessment of normal distribution; exploring data and descriptive statistics; effect sizes and power calculations; parametric and non-parametric tests comparing the differences between same subject / matched designs and different subject designs; repeated measures analysis of variance and analysis of covariance; tests of association between variables including correlation, regression, logistic regression and multiple regression; introduction to factorial analysis including principle component analysis.

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Things that must be done this week

- **Online Enrolment**

- You must complete Online Enrolment. After enrolment you will have access to the University Digital Learning Environment (DLE) and your timetable etc.
- To complete online enrolment you will need to make your module choices.

- **Medical information**
- As soon as you have completed online enrolment, please complete the online medical questionnaire at the following link (go to 'preview medical and travel form' on the left hand menu), https://s3.plymouth.ac.uk/s3/
- The information is required solely to enable organisers of field courses, lab work and other activities to be alerted to any special requirements or precautions which may be necessary.

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When / How I choose my dissertation ?

- Semester 1: Potential project titles and supervisors, November / December
- Semester 2: Start project work alongside taught modules
- Summer: Full-time research, submission of final dissertation mid-September.
- Computing: 24 hour computing facilities available on campus

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Student Representative(s)

You need to select one or two representatives

- Contact point
- Committee member

If you are interested and would like to nominate yourself as Course Rep, please email me by Monday the 5th October with a short paragraph explaining why students should vote for you.

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